

MODEL VERSION SELECT

Clean Dataset - Train Set : S1 (5000), S2 (50000)

Real Dataset - Test Set : S3 (55000), S4 (105000)

Pair	Version	Features	Clean Accuracy	Real Accuracy	Gap	Real Predict Runtime	Pipeline Runtime
S1 5,000 vs S3 55,000	V1	25	80.60%	80.09%	-0.51 pp	1.5294s	0.6297s
S1 5,000 vs S3 55,000	V2	64	81.20%	80.71%	-0.49 pp	0.8747s	30.5754s
S1 5,000 vs S3 55,000	V3	81	81.90%	82.04%	+0.14 pp	0.6700s	33.0476s
S1 5,000 vs S3 55,000	V4	107	82.50%	82.05%	-0.45 pp	1.0841s	33.4670s
S1 5,000 vs S3 55,000	V5	64	82.00%	81.91%	-0.09 pp	0.6660s	33.4400s
S2 50,000 vs S4 105,000	V1	25	82.29%	82.45%	+0.16 pp	0.7325s	2.1396s
S2 50,000 vs S4 105,000	V2	64	82.17%	81.92%	-0.25 pp	1.0347s	230.2552s
S2 50,000 vs S4 105,000	V3	81	83.04%	82.87%	-0.17 pp	1.0500s	254.8428s
S2 50,000 vs S4 105,000	V4	107	83.16%	82.86%	-0.30 pp	1.2532s	258.5415s
S2 50,000 vs S4 105,000	V5	64	82.86%	82.78%	-0.08 pp	0.9904s	258.4398s

S1-like DB 5,000 rows

Version	Features	Query/Load	Clean	Feature Eng.	Load Schema	Load Model	Predict	Total
V1	25	0.4334s	0.1230s	0.1308s	0.0017s	0.0101s	0.0537s	0.6297s
V2	64	0.4334s	0.1230s	30.0664s	0.0014s	0.0109s	0.0633s	30.5754s
V3	81	0.4334s	0.1230s	32.5470s	0.0016s	0.0079s	0.0577s	33.0476s
V4	107	0.4334s	0.1230s	32.9234s	0.0020s	0.0064s	0.1017s	33.4670s
V5	64	0.4334s	0.1230s	32.9269s	0.0019s	0.0070s	0.0708s	33.4400s

S2-like DB 50,000 rows

Version	Features	Query/Load	Clean	Feature Eng.	Load Schema	Load Model	Predict	Total
V1	25	1.0301s	0.7077s	0.7411s	0.0104s	0.0246s	0.3334s	2.1396s
V2	64	1.0301s	0.7077s	228.6391s	0.0021s	0.0163s	0.5675s	230.2552s
V3	81	1.0301s	0.7077s	253.3153s	0.0033s	0.0144s	0.4796s	254.8428s
V4	107	1.0301s	0.7077s	256.8114s	0.0037s	0.0113s	0.6849s	258.5415s
V5	64	1.0301s	0.7077s	256.8475s	0.0030s	0.0138s	0.5453s	258.4398s



ลูกค้า : C000722

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order no	order_id	days back	order_date	category	brand	payment	channel	amount	returned	reason
10	LGS1ORD0002729	177	2025-08-05	Fashion	UrbanWear	Credit_Card	Shopee	787.94	Returned	Not as Expected
11	LGS1ORD0002882	165	2025-08-17	Health	MediCare	Credit_Card	Shopee	4334.52	Not Returned	Not Returned
12	LGS1ORD0003517	115	2025-10-06	Electronics	GadgetWorld	COD	Web	1412.41	Returned	Not as Expected
13	LGS1ORD0004206	60	2025-11-30	Fashion	UrbanWear	Credit_Card	TV_Show	585.88	Not Returned	Not Returned
14	LGS1ORD0004436	42	2025-12-18	Home_Appliance	CleanMate	COD	Shopee	2676.22	Returned	Damaged Packaging
15	LGS1ORD0004473	39	2025-12-21	Electronics	GadgetWorld	Credit_Card	CallCenter	6344.81	Not Returned	Not Returned

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order no	order_id	days back	order_date	category	brand	payment	channel	amount	returned	reason
1	LGS1ORD0000369	364	2025-01-30	Kitchen	CookEasy	COD	Shopee	8403.92	Not Returned	Not Returned
2	LGS1ORD0000783	331	2025-03-03	Kitchen	KitchenMax	PromptPay	Shopee	16961.44	Not Returned	Not Returned
3	LGS1ORD0000894	322	2025-03-12	Electronics	NovaTech	PromptPay	Shopee	16267.30	Not Returned	Not Returned
4	LGS1ORD0000944	318	2025-03-16	Home_Appliance	CleanMate	Credit_Card	Shopee	8474.70	Not Returned	Not Returned
5	LGS1ORD0001244	295	2025-04-09	Electronics	ShieldCase	COD	Lazada	5251.54	Returned	Not as Expected
6	LGS1ORD0001480	276	2025-04-28	Health	WellPlus	COD	TV_Show	849.72	Not Returned	Not Returned
7	LGS1ORD0001804	250	2025-05-23	Fashion	UrbanWear	COD	Web	924.62	Returned	Size Issue
8	LGS1ORD0001890	244	2025-05-30	Fashion	FitStyle	Credit_Card	Shopee	1622.72	Returned	Changed Mind
9	LGS1ORD0002679	181	2025-08-01	Health	WellPlus	COD	CallCenter	7394.44	Not Returned	Not Returned
10	LGS1ORD0002729	177	2025-08-05	Fashion	UrbanWear	Credit_Card	Shopee	787.94	Returned	Not as Expected
11	LGS1ORD0002882	165	2025-08-17	Health	MediCare	Credit_Card	Shopee	4334.52	Not Returned	Not Returned
12	LGS1ORD0003517	115	2025-10-06	Electronics	GadgetWorld	COD	Web	1412.41	Returned	Not as Expected
13	LGS1ORD0004206	60	2025-11-30	Fashion	UrbanWear	Credit_Card	TV_Show	585.88	Not Returned	Not Returned

order no	order_id	days back	order_date	category	brand	payment	channel	amount	returned	reason
14	LGS1ORD0004436	42	2025-12-18	Home_Appliance	CleanMate	COD	Shopee	2676.22	Returned	Damaged Packaging
15	LGS1ORD0004473	39	2025-12-21	Electronics	GadgetWorld	Credit_Card	CallCenter	6344.81	Not Returned	Not Returned

Customer Query Just 1 row 1 Person

Lookback	จำนวน rows ที่ดึงได้	คืนสินค้า	ไม่คืน	Return Rate	Runtime
6 เดือน / 180 วัน	6 rows	3	3	50.00%	1.2659 ms
1 ปี / 365 วัน	15 rows	6	9	40.00%	1.2327 ms

Clean Dataset S1 vs S2

Version	Features	S1 Accuracy	S2 Accuracy	Gap	S1 Predict	S2 Predict	S1 Pipeline	S2 Pipeline
V1	25	80.60%	82.29%	+1.69 pp	0.1390s	0.3488s	0.6297s	2.1396s
V2	64	81.20%	82.17%	+0.97 pp	0.0795s	0.4927s	30.5754s	230.2552s
V3	81	81.90%	83.04%	+1.14 pp	0.0609s	0.5000s	33.0476s	254.8428s
V4	107	82.50%	83.16%	+0.66 pp	0.0986s	0.5968s	33.4670s	258.5415s
V5	64	82.00%	82.86%	+0.86 pp	0.0605s	0.4716s	33.4400s	258.4398s

Real Dataset S3 vs S4

Version	Features	S3 Accuracy	S4 Accuracy	Gap	S3 Predict	S4 Predict	S3 Pipeline	S4 Pipeline
V1	25	80.09%	82.45%	+2.36 pp	1.5294s	0.7325s	2.3536s	4.4932s
V2	64	80.71%	81.92%	+1.21 pp	0.8747s	1.0347s	253.2807s	483.5358s
V3	81	82.04%	82.87%	+0.83 pp	0.6700s	1.0500s	280.3271s	535.1699s
V4	107	82.05%	82.86%	+0.81 pp	1.0841s	1.2532s	284.3956s	542.9370s
V5	64	81.91%	82.78%	+0.87 pp	0.6660s	0.9904s	284.2838s	542.7235s

สาเหตุที่ Run-Time บาง Version ใช้เวลานานกว่าเป็นเพราะ :

1. การ Join หลาย Table เช่น table customer, table product
2. คำนวน Customer History (ประวัติลูกค้า) รายการ/order เมื่อลูกค้ามีหลายรายการ
3. คำนวน ประวัติลูกค้าย้อนหลัง เช่น 7/30/90/365 วัน
4. คำนวน return rate ราย product/category/courier/payment
5. จำนวน Feature ที่เพิ่มมาทำให้ model ทำงานหนักขึ้น

Version	Features	Clean S1 5,000	Real S3 55,000	Gap S3-S1	Clean S2 50,000	Real S4 105,000	Gap S4-S2
V1	25	80.60%	80.09%	-0.51 pp	82.29%	82.45%	+0.16 pp
V2	64	81.20%	80.71%	-0.49 pp	82.17%	81.92%	-0.25 pp
V3	81	81.90%	82.04%	+0.14 pp	83.04%	82.87%	-0.17 pp
V4	107	82.50%	82.05%	-0.45 pp	83.16%	82.86%	-0.30 pp
V5	64	82.00%	81.91%	-0.09 pp	82.86%	82.78%	-0.08 pp